🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 · STANDARD 6.NOS.4

Prime Factorization Flagship

Lesson 1-1-flagship · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

ORBITAL LOGISTICS MISSION

Cargo Codebreak

You are the logistics officer aboard Station Helios. A shipment of 60 supply crates just docked, and the sorting robots can only distribute cargo once it is broken into its prime building blocks. Master prime factorization and the station eats this week.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write a number as a product of its prime factors using a factor tree.

Language Objective: I can explain how I broke a number down using the words prime number, composite number, factor, and exponent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Prime number | **Composite number** | **Prime factorization** | **Factor tree** | **Exponent**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A whole number greater than 1 with exactly two factors, 1 and itself, is a ____________ number.

2. A whole number greater than 1 that has more than two factors is a ____________ number.

3. ____________ — Writing a number as prime numbers multiplied together.

4. I can break a number into its prime factors step by step using a ____________.

5 In 2^3, the small number 3 is the ______________ and it shows 2 is multiplied 3 times.

Write About the Math **Guided ESOL writing**

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Look at your factor tree for 48. How did you decide which numbers to break apart first, and how did you know when to stop?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Prime number** — A number bigger than 1 that you can only divide by 1 and itself.
  Número primo — Un número mayor que 1 que solo se puede dividir entre 1 y sí mismo.
- [ ] **Composite number** — A number bigger than 1 that you can divide by more than just 1 and itself.
  Número compuesto — Un número mayor que 1 que se puede dividir entre más números, no solo 1 y sí mismo.
- [ ] **Prime factorization** — Writing a number as prime numbers multiplied together.
  Factorización prima — Escribir un número como números primos multiplicados.
- [ ] **Factor tree** — A picture that splits a number into its prime numbers, step by step.
  Árbol de factores — Un dibujo que separa un número en sus números primos, paso a paso.
- [ ] **Exponent** — A small number that tells how many times to multiply a number by itself.
  Exponente — Un número pequeño que dice cuántas veces multiplicar un número por sí mismo.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

I broke 48 into ____ first because ____.
Separé 48 en ____ primero porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 48 is composite because it has more than two factors.

Evidence: Students say 48 is composite because it has more than two factors (divisible by 2, 3, 4, 6, ...), so it can be broken into smaller groups.

Reasoning: This evidence connects the definition of prime factorization to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I broke 48 into ____ first because ____.**
  Separé 48 en ____ primero porque ____.
- **I knew I was finished when every factor was ____.**
  Supe que terminé cuando cada factor era ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Prime number
2. **Notes 2:** Composite number
3. **Notes 3:** Prime factorization
4. **Notes 4:** Factor tree
5. **Notes 5:** Exponent
6. **Watch for this mistake:** *A common mistake in Prime Factorization is stopping the factor tree too soon — leaving a composite number like 4, 6, 8, or 9 in the final answer instead of continuing to split it. For example, writing $20 = 4 \times 5$ as the prime factorization (4 is composite) instead of finishing with $20 = 2 \times 2 \times 5$. Before you submit, check every number in your final answer: if any factor can still be broken into smaller factors, the tree isn't finished yet.*
7. **Try It 1:** A. $2^3 \times 3^2$ — *$72 = 2 \times 2 \times 2 \times 3 \times 3$. Three 2s and two 3s give $2^3 \times 3^2$.*
8. **Try It 2:** A. $2 \times 2 \times 3 \times 7$ — *$84 = 4 \times 21 = (2 \times 2) \times (3 \times 7) = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$. The vault opens!*